

1. Information about the indicator	
Goal	End hunger, achieve food security and improved nutrition and promote sustainable agriculture
Task	By 2030, end hunger and ensure access by all people, in particular the poor and people in vulnerable situations, including infants, to safe, nutritious and sufficient food all year round
The indicator	Prevalence of moderate or severe food insecurity in the population, based on the Food Insecurity Experience Scale (FIES)
2. Information about the organization	
Organization	Bureau of National Statistics of the Agency for Strategic Planning and Reforms of the Republic of Kazakhstan
3. Definitions and concepts	
Definition	The indicator measures the percentage of people in the population who experienced moderate or severe food shortages during the reporting period. The severity of food insecurity, defined as a hidden feature, is measured by the global benchmark scale of food security experience, according to the measurement standard created by FAO through the application of the Food Security Experience Scale (hereinafter referred to as SHOPB) in more than 140 countries around the world, starting in 2014.
Basic concepts:	Extensive research over more than 25 years has shown that the inability to access food leads to a number of events and conditions that are quite common in different cultures and socio-economic conditions, ranging from the task of getting enough food, to the need to compromise on the quality or variety of food consumed, to forced reduction. food consumption by reducing portion sizes or skipping meals, to the point of extreme hunger and lack of means of access to food. Typical conditions like these form the basis for building scales based on experience in measuring food insecurity. When analyzed using reliable statistical methods based on the Theory of Response to individual articles, data collected at such a scale provides the basis for calculating theoretically consistent, country-by-country food insecurity prevalence rates. The severity of food insecurity measured by this indicator therefore directly reflects the extent to which households or individuals are unable to regularly access the food they need.
Justification and interpretation	Food insecurity at a moderate level of severity is usually associated with the inability to regularly receive a healthy, balanced diet. Thus, the high prevalence of food insecurity at moderate levels can be considered as a prediction of various forms of population health conditions related to nutrition and micronutrient deficiencies and unbalanced nutrition. On the other hand, severe levels of food insecurity imply a high probability of reduced food intake and, therefore, can lead to more serious forms of malnutrition, including hunger. Short questionnaires like SHOPB are very easy to administer at limited cost, which is one of the main advantages of using them. However, the ability to accurately determine the food insecurity status of specific individuals or households is limited by a small number of questions, so it

	is advisable to assign individual respondents questions on individual food insecurity classes, which is best done in probabilistic conditions. Thus, ensuring that estimates of the prevalence of the indicator in the population are sufficiently reliable, even if they are based on relatively small sample sizes. As with any statistical assessment, reliability and accuracy depend crucially on the quality of survey design and implementation. One of the main advantages of analytical data processing using Rasch model-based methods is that they allow you to verify the quality of the collected data and assess the likely margin of uncertainty around estimated prevalence rates, which should always be reported.
4. Data sources and collection methods	
Data sources Data	Sample survey of households by standard of living.
Collection Methods	Sample survey of households by standard of living.
Unit of measurement	Percent
Periodicity	Annual
5. Calculation method and other methodological considerations	
Calculation method:	Data at the individual or household level can be collected using one of several food safety questionnaires based on experience. Each of these survey modules collects answers to questions that require a report on the occurrence of a number of typical events and conditions related to food insecurity. The data can be analyzed using the Rasch model (also known as the 1-PL one-parameter logistic model), which sets the condition that the probability of an affirmative answer by respondent i to question j is a logistic function of the distance, based on severity, between the respondent's position and the subject's position. $\text{Prob}\{X_{i,j} = \text{Yes}\} = (\exp(a_i - b_j)) / (1 + \exp(a_i - b_j))$ The parameters a_i and b_j can be estimated using maximum likelihood procedures. In particular, the a_i parameters are interpreted as a measure of the severity of the food insecurity for each respondent and are used to classify them into food insecurity classes. SHOPB considers three classes: (a) food insecurity or moderate food insecurity; (b) moderate or severe food insecurity; and (c) severe food shortages and an assessment of the probability of moderate or severe food shortages ($p_{\text{mod} + \text{sev}}$) and the probability that there is a severe food shortage for each respondent (p_{sev}), from $0 < p_{\text{sev}} < p_{\text{mod} + \text{sev}} < 1$. The probability that the food is safe or there is moderate food insecurity can be obtained as $p_{\text{fs}} = 1 - p_{\text{mod} + \text{sev}}$.

Taking into account a representative sample, the prevalence of food insecurity at moderate or severe levels ($FI_{mod+sev}$) and at severe levels (FI_{sev}) in a population is calculated as a weighted sum of the probability of belonging to a group of moderate or severe food insecurity, as well as to a serious class of food insecurity, respectively, for all individual or household respondents in the sample:

$$FI_{mod+sev} = \sum p_{i_{mod+sev}} \times w_i$$

and also

$$FI_{sev} = \sum p_{i_{sev}} \times w_i$$

Where w_i - are the weighting coefficients after stratification, which indicate the proportion of individuals or households in the national population represented by each element in the sample.

It is important to note that if w_i are the individual weights of the sample, then the prevalence of food insecurity refers to the total population (individuals), and if they are the weight of households, then the prevalence refers to the number of households. To calculate indicator 2.1.2, the aim is to determine the prevalence of food insecurity among individuals. This implies the following:

If the survey is conducted at the household level and provides household weights, then they should be converted to individual weights by multiplying the weights by the size (number) of the household. This individual weighing system can then be used to calculate individual food insecurity prevalence rates in formulas (1) and (2).

If the survey includes only adults, then the adult weights applied to the probabilities in formulas (1) and (2) provide the adult prevalence rate. (FI_{Adults}). In this case, in order to calculate the prevalence in the total population, it is also necessary to calculate the proportion of children who live in households where at least one adult is food insecure. This can be done by dividing the weight of adults by the number of adults in the household and multiplying these approximate weights by the number of children in the family.

After obtaining approximate child weights, the prevalence of food insecurity among children who live in households where at least one adult is food insecure ($FI_{Children}$) can be calculated by applying these weights to the probabilities of food insecurity in formulas (1) and (2). The prevalence of food insecurity in the total population is finally calculated as:

$$FI_{mod+sev} = \frac{FI_{Adults\ mod+sev} * N_{Adults} + FI_{Children\ mod+sev} * N_{Children}}{N_{Adults} + N_{Children}}$$

And also

$$FI_{sev} = \frac{FI_{Adults\ sev} * N_{Adults} + FI_{Children\ sev} * N_{Children}}{N_{Adults} + N_{Children}}$$

	Where N_{Adults} and $N_{Children}$ are the adult and child populations in the country. When applied to the total population of a country, the prevalence of food insecurity in the total population shows the number of people who live in food-insecure homes (or in households where at least one adult is food insecure) in a country at varying levels of severity ($N_{mod+sev}$ and N_{sev}). In the database, the number of people without food insecurity is expressed in thousands.
Comments and restrictions:	
6. Data availability and disaggregation	
Data availability and gaps:	The data is available on the website of the Bureau of National Statistics of the Agency for Strategic Planning and Reforms of the Republic of Kazakhstan.
The level of disaggregation:	-
7. Comparability with international data/standards	-
8. References and documentation	www.stat.gov.kz